

# **OmniGuard: Multimodal AI-Based Authenticity and Fraud Detection System**

## **Problem Statement**

In today's digital world, people are constantly exposed to content such as images, videos, voice recordings, and text messages through social media, messaging platforms, emails, and online advertisements. With the rapid growth of Artificial Intelligence, it has become increasingly easy to create highly realistic fake content, including deepfake videos, cloned voices, forged documents, AI-generated images, and misleading text. These manipulated files are often used in financial scams, fake job offers, phishing attacks, deceptive product promotions, and the spread of misinformation. As a result, many users find it difficult to determine whether the content they receive is genuine or fabricated, which can lead to financial loss, identity theft, and poor decision-making.

The proposed solution, OmniGuard, is a multimodal AI-powered web application designed to help users verify the authenticity of digital content. The system allows users to upload text, images, audio recordings, and video files through a simple drag-and-drop interface. Once the files are uploaded, the application analyzes each type of data using machine learning and deep learning techniques. Text is examined to identify phishing messages, scam emails, fake reviews, and misleading claims. Images are analyzed to detect edited photos, forged documents, manipulated screenshots, and AI-generated visuals. Audio recordings are processed to determine whether the voice is natural, cloned, or synthetically generated. Videos are inspected to detect deepfakes, facial manipulations, and AI-generated testimonials. After analyzing all uploaded content, the system generates an overall trust score and provides a detailed report indicating whether the content is safe, suspicious, or likely fraudulent, along with recommendations to guide the user.

OmniGuard aims to serve as a universal authenticity verification platform that can be used by students, professionals, families, businesses, and senior citizens. The project combines Natural Language Processing, Computer Vision, Audio Signal Processing, Multimodal Machine Learning, Full-Stack Web Development, and MLOps into a single practical application with strong social impact. By helping users distinguish between real and fake digital content, the system can reduce fraud, improve online safety, and build trust in digital communication.